

Music Player App Development Roadmap (Kotlin, Android)

Week 1: Setting Up and Learning Basics

Goals:

- Set up Android Studio and configure a Kotlin project.
- Learn the basics of Kotlin (variables, functions, classes, and OOP concepts).
- Understand Android app components (Activities, Fragments, and Intents).
- Create a basic UI with Jetpack Compose or XML.

Tasks:

- Install Android Studio and set up Kotlin.
- Follow an introductory Kotlin course or tutorial.
- Build a simple “Hello World” Android app.
- Learn about Activity and Fragment.

Resources:

- Kotlin Basics: [Kotlin Docs](#)
 - Android Fundamentals: [Google Android Developer Guide](#)
 - YouTube: [Kotlin for Beginners](#)
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Week 2: UI Design and Navigation

Goals:

- Learn **Jetpack Compose** (recommended) or XML for UI design.
- Implement **Navigation Component** for switching between screens.
- Create a **basic UI** (Home Screen, Search Screen, Library Screen).
- Learn how to handle **RecyclerView/ListView** for displaying songs.

Tasks:

- Design screens for searching, playing, and saving music.
- Implement a bottom navigation bar (Home, Search, Library).
- Fetch and display a list of dummy songs.

Resources:

- Jetpack Compose: [Jetpack Compose Basics](#)
 - RecyclerView: [RecyclerView Guide](#)
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Week 3: Implementing Audio Playback

Goals:

- Learn how to use **ExoPlayer** for audio playback.
- Implement **basic music playback** (Play, Pause, Stop).
- Handle **media controls** (seek bar, play/pause button).

Tasks:

- Integrate ExoPlayer and play a sample audio file.
- Implement media controls (Play, Pause, Seek).
- Display the currently playing song.

Resources:

- ExoPlayer Setup: [ExoPlayer Guide](#)
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Week 4: Searching and Streaming Music Online

Goals:

- Learn how to **fetch online songs** using APIs (YouTube API or any free music API).
- Implement **search functionality**.

Tasks:

- Integrate an API to fetch songs.
- Show results in a list and allow users to play them.

Resources:

- YouTube API: [YouTube API Docs](#)
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Week 5: Implementing Video Playback & Background Play

Goals:

- Learn **ExoPlayer's video playback**.
- Implement **switching between video and audio mode**.
- Handle **background playback** when the app is minimized or the screen is turned off.

Tasks:

- Add video player support.
- Implement an automatic switch from video to audio when the screen is off.

Resources:

- Background Playback: [Foreground Services](#)
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Week 6: User Library and Storage

Goals:

- Learn **Room Database** for storing user data.
- Implement **saving favorite songs and playlists**.
- Support **offline playback** by accessing downloaded music from storage.

Tasks:

- Allow users to save their favorite songs to a database.
- Implement functionality to load and play local songs.

Resources:

- Room Database: [Room Persistence Library](#)
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Week 7: Enhancements and Smart Recommendations

Goals:

- Implement **music recommendations** based on user activity.
- Optimize UI/UX.
- Add **persistent notifications** for playback.

Tasks:

- Suggest music based on past searches and plays.
- Show a **persistent music player notification** when a song is playing.

Resources:

- Notifications: [Android Notification Guide](#)
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Week 8: Testing, Bug Fixes, and Deployment

Goals:

- Test the app for bugs.
- Optimize performance.
- Deploy the app on the Play Store (optional).

Tasks:

- Run thorough testing.

- Fix UI issues and crashes.
- (Optional) Learn how to publish an app on the Play Store.

Resources:

- Play Store Deployment: [Google Play Console](#)

This roadmap will help you **learn Kotlin and Android development while building your app step by step.**